

Mirroring the Scene Flips the Score: Directional Success Depends on Layout in Language-Conditioned Manipulation

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Abstract—Directional success gaps between opposite spatial commands are often read as evidence of language grounding, although completion also depends on the scene in which a command is executed. We test that attribution with matched interventions that hold the checkpoint, controller, and seeds fixed while either reflecting object layout with the prompts unchanged or re-expressing the same physical goals in another linguistic reference frame. For $\pi_{0.5}$, reflecting only the movable-object centers reverses completion: 4/27 LEFT and 25/27 RIGHT become 25/27 and 9/27. The same checkpoint therefore favors opposite directions in two layouts; its directional gap is not a checkpoint-level grounding property. On matched physical goals, expressing the relation from the other object’s frame collapses $\pi_{0.5}$ endpoint separation from 23.28 cm to 0.19 cm. Nano’s completion interaction is zero, yet matched requested-side depth shifts by 13.9 cm. A 20.2-percentage-point $\pi_{0.5}$ gap remains in the symmetric scene and does not meet the prespecified equivalence criterion; the control changes the gap without identifying its cause. Directional success is therefore a property of the policy–layout pair and cannot, by itself, isolate language grounding.

Index Terms—Vision-language-action models, robot learning, manipulation, language grounding, evaluation methodology.

I. INTRODUCTION

Vision-language-action (VLA) policies promise a single interface for open-ended robot control: a user states a goal in words, and one pretrained model produces the actions [1]–[4]. Whether that interface delivers is an empirical question, and relational placement has become a standard way to ask it. A representative protocol holds the scene fixed and asks the policy to place an object in “the pot on the left” or “the pot on the right.” It balances trials between the two directions and scores each with a frozen success rubric [5]. It is tempting to read a large LEFT/RIGHT gap as a language-grounding failure: the model understood one direction but not the other, or vision overrode language [6].

That reading is plausible because robot datasets often use formulaic, low-diversity instruction labels, making the command predictable from the observation and allowing a conditioned policy to ignore language [7]–[9]. The attribution question is whether a binary directional score demonstrates that failure in a particular experiment [10].

That score combines several questions. For a trial to succeed, the instruction has to alter control, the change has to point

toward the requested relation, the robot has to execute a feasible grasp and transport, and the final state has to cross a binary task boundary. A policy can pass the first two stages yet fail the last two because, for that checkpoint, one requested placement is harder to complete in the sampled configuration. The reverse mistake is also possible: a policy can react to the literal tokens LEFT and RIGHT without preserving the physical goal once the linguistic reference frame is inverted [11]. A single binary contrast can therefore produce two opposite errors: a false failure when language redirects the endpoint but scene-conditioned task competence differs by direction, and a false pass when familiar wording elicits the expected behavior without relation-level transfer.

Our central finding is an invariance failure: the directional gap is a property of the policy–layout pair, not a layout-invariant property of the checkpoint. Reflection reverses completion with checkpoint, prompts, controller, and seeds fixed (Fig. 1); Sec. IV gives the paired analysis.

Scene dependence is not the only threat to interpretation. When the same physical goals are expressed from the other object’s reference frame—“cube left of bowl” becomes “bowl right of cube,” and conversely—the familiar base-frame endpoint separation nearly vanishes. Base-frame sensitivity to LEFT and RIGHT therefore does not establish that the underlying relation transfers across linguistic frames.

The controls change what a directional score can support. A large gap can be a false grounding failure when one layout makes one requested placement harder to complete; a reassuring base-frame response can be a false pass when it does not survive the tested equivalent expression of the relation. Reflection exposes the first error, reference inversion exposes the second, and the endpoint decomposition makes both visible.

We make three contributions, ordered by the claims they support:

- **A matched intervention that identifies layout dependence.** A pure object-position reflection supplies the decisive within-checkpoint comparison; graded displacement and symmetric-scene controls test the same dependence in continuous placement.
- **A reference-inversion control for relation-level transfer.** Holding each physical goal and the manipulated object fixed while changing the linguistic frame shows that familiar base-frame endpoint separation does not survive the tested reference inversion.

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- **A decomposition that separates redirection from completion.** Executed-action response, signed endpoint ordering, requested-side depth, and frozen task completion show where a directional contrast changes. This decomposition exposes layout-dependent placement even when binary success is at ceiling.

None requires retraining, new hardware, or a change to the frozen success predicate.

II. RELATED WORK

Generalist language-conditioned policies. Modern manipulation policies pair pretrained visual-language representations with expressive action decoders. Representative systems include π_0 and $\pi_{0.5}$, which use flow-matching action models, Open-VLA, an open autoregressive alternative, and RT-2, which transfers web knowledge to control [1]–[4]. Hierarchical variants place a reasoning model above the controller and pass it natural-language subtasks [12]. Scale has expanded the tasks one policy can address, but it also makes a failed episode difficult to attribute: language interpretation, object localization, grasping, transport, release, and task scoring all lie on the same path to success.

Steerability and instruction following. A growing literature treats weak instruction following as the bottleneck and enriches the command interface. CAST augments datasets with counterfactual language [13], while STEER relabels demonstrations to expose a low-level control interface [14]. Steerable Policies train on subtask, motion, trace, and point commands [5], and InSight makes individual primitives addressable [15]. Other work steers a frozen policy at test time [16] or asks how steerability survives post-training [10], [17]. These methods address genuine failures of instruction following, but they rely on directional and relational success contrasts to certify that the failure was linguistic in the first place.

Diagnosing whether language is used. Counterfactual evaluation reduces attribution ambiguity by holding the visual input fixed while changing the instruction. LIBERO-CF and Counterfactual Action Guidance examine cases in which visual evidence dominates a conflicting instruction [6], and recent analyses bound how far instruction following transfers to unseen phrasings [11]. Steering protocols also report graded task progression, per-primitive progress, or success by perturbation condition [5], [15], [16]. Those approaches grade progress toward the command that was issued. Our contrast asks a different question: when two commands share the same reset and policy-sampling seed, what changes because one relation word changes? The exact pairing requires a resettable simulator, but it prevents scene and sampling differences from entering the prompt contrast [15].

Evaluation practice. Grounding spatial and directional language into executable behavior long predates VLAs, and early systems already separated the referring expression, the spatial relation, and the executed path [18], [19]. Recent large-scale evaluations, including the TRI study of Large Behavior Models, strengthen manipulation comparisons through controlled initial

conditions, randomized trials, and statistical confidence [20]. We share that emphasis on controlled evaluation but target a different estimand: rather than aggregate success or robustness across tasks, we contrast two relational commands at the same reset and sampling seed. DROID and RoboTwin illustrate the distinct data and simulation ecosystems in which such policies are trained and tested [21], [22]. A frozen binary predicate answers whether the task was completed in the evaluated scene distribution, but it can saturate at ceiling, collapse at floor, or hide large continuous placement changes. We keep it as the final outcome and pair it with signed endpoint and requested-depth measurements and with interventions that vary language form and scene geometry separately.

III. MATCHED AUDIT DESIGN

The audit layers two controls on the same base-frame prompt pair. Reference inversion changes the linguistic frame while holding each physical goal fixed; the scene interventions change object geometry while preserving the prompt pair. Together they test whether a directional score is stable to either representation of the task.

Minimal-pair token intervention. Literal prompt tokens appear in small capitals, LEFT and RIGHT, while *leftward* and *rightward* denote physical directions in robot coordinates; a policy can track the token without preserving the direction it denotes. For each reset s_i we run a policy under prompts ℓ_i^L and ℓ_i^R that differ only in one opposite direction word, holding scene state, checkpoint, controller, environment seed, policy-sampling seed, action interface, and horizon fixed. Both prompts therefore execute from the same initial state and random draws; the direction word is the designed difference. We issue the instruction once at reset and leave it alone: no oracle actions, no prompt switching, and no language conditioned on progress. DROID/RoboLab pairs the DROID Franka embodiment [21] with the RoboLab Isaac Sim environment [23] and uses a Rubik’s cube and bowl; RoboTwin [22] uses a wooden block and playing-card box. Both require placement and release in the requested relational region.

Checkpoints and pre-intervention screen. Table I lists the eight expanded Phase-A rows; App. B records the available artifact identifier and revision behind each one. Two are action-only VLAs and six are world-action models (WAM), which predict future observations alongside actions. Within each arena, checkpoints use its fixed action interface and frozen predicate. Edge and Nano are the 4B and 16B policies of one Cosmos3 family [24], so the rows are not fully independent. The checkpoints come from different developers with different pretraining mixtures and post-training recipes; exact training-episode membership, caption exposure, and sampling weights are undisclosed. Every causal contrast is therefore within checkpoint. Cross-row differences provide context only; no comparison across rows is load-bearing.

We define the pre-intervention screen as one completed matched-pair row per checkpoint and arena. Repository evidence spans eleven checkpoint identities. Of these, the Phase-A

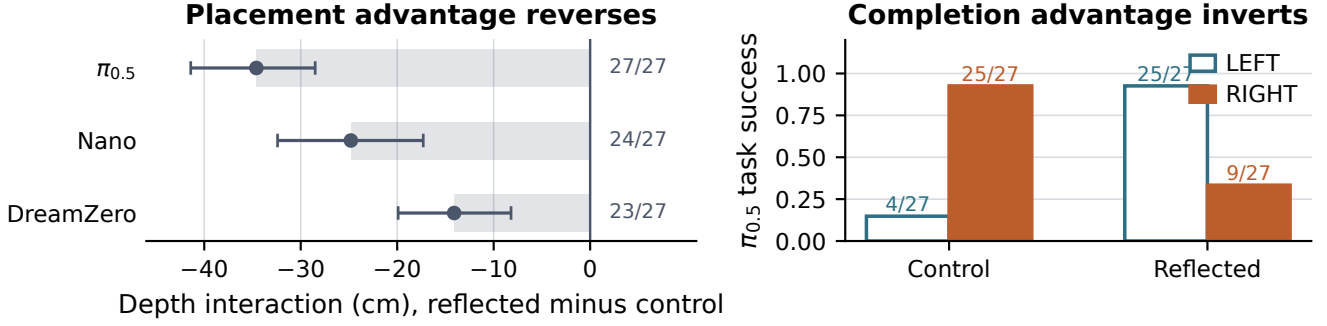


Fig. 1. **Reflection changes continuous placement and reverses $\pi_{0.5}$ completion.** Left: reflected-minus-control interaction in requested-side depth, with seed-clustered 95% bootstrap intervals; fractions give seeds with a negative interaction. Right: the $\pi_{0.5}$ completion advantage reverses after reflection; plotted fractions give the success counts. All causal comparisons are within checkpoint; the panels do not support cross-model ranking.

TABLE I
EXPANDED PHASE-A CHECKPOINT-ARENA ROWS IN THE
PRE-INTERVENTION SCREEN; SHORT NAMES IN PARENTHESES.

Model (short name)	Class	Pairs
<i>DROID/RoboLab: cube to the requested side of a bowl</i>		
$\pi_{0.5}$ DROID [4]	VLA	27
GR00T N1.7 DROID (GR00T) [25]	VLA	27
Cosmos3 Edge Policy DROID (Edge) [24]	WAM	27
Cosmos3 Nano Policy DROID (Nano) [24]	WAM	27
DreamZero DROID (DreamZero) [26]	WAM	27
<i>RoboTwin: block to the requested side of a card box</i>		
Efficient-WAM-RT [27]	WAM	63
FastWAM [28]	WAM	63
LingBot-VA [29]	WAM	63

screen reported here comprises eight checkpoint-arena rows: five DROID/RoboLab rows at 27 matched pairs each and three RoboTwin rows at 63 matched pairs each, for 324 pairs or 648 episodes. The remaining identities appear only in separate cohorts: a 20-pair π_0 -FAST compatibility bridge and two bounded three-pair RoboTwin gates, LingBot-VLA 4B (distinct from the screened LingBot-VA checkpoint) and Light-WAM. Because those cohorts use different designs and denominators, we do not pool them into the Phase-A screen. The two arenas use different tasks and different success tests, so no outcome estimate or effect pools arenas; totals across rows are bookkeeping only, and the larger intervention cohorts state their own denominators.

Behavioral measurements. Let $a_{1:T_L}^L$ and $a_{1:T_R}^R$ be the executed action sequences in a minimal pair, and let y_i^L, y_i^R be the final target-minus-reference lateral offsets in robot coordinates, where positive is robot-left. The four measurements correspond to four stages of the task: prompt response, directional ordering, placement quality, and completion. We report

$$A_i = \mathbb{I}[a_{1:T}^L \neq a_{1:T}^R], \quad T = \min(T_L, T_R), \quad (1)$$

$$D_i = y_i^L - y_i^R, \quad (2)$$

$$S_i^d = \mathbb{I}[\text{frozen task predicate is satisfied for } d], \quad (3)$$

$$M_i^L = y_i^L, \quad M_i^R = -y_i^R. \quad (4)$$

A_i uses bitwise inequality on the complete common executed prefix. Positive D_i means the requested endpoint ordering: within a pair the object came to rest further leftward under LEFT than under RIGHT. S_i^d is task completion, and requested-side depth M_i^d records where the target comes to rest along the requested physical direction.

Calibrating A_i against sampling noise. A_i tests response, not semantics, and alone it is weak evidence: stochastic sampling makes two executions differ even under an unchanged prompt. We therefore calibrate it on a single frozen observation, querying the policy without executing the robot so that no difference compounds through dynamics. The diagnostic issues 336 action requests across six checkpoint-layout conditions (three DROID/RoboLab checkpoints crossed with the control and reflected scenes). Each condition contains 27 matched LEFT/RIGHT seed pairs and one exact repeat per direction at seed 9400; all 12 repeat comparisons are bit-identical. For seed s , prompt effect is $\text{RMS}(a_s^L, a_s^R)$ over the complete native returned action arrays; the noise pool contains $\text{RMS}(a_s^d, a_{s'}^d)$ for each prompt d and every distinct-seed pair $s < s'$. The prompt-to-noise ratio divides the median of the 27 prompt effects by the median of the 702 pooled same-prompt distances. The exact repeats verify reproducibility and do not define the noise denominator; when the pooled cross-seed median is zero, the ratio is undefined and we report the prompt-effect numerator alone.

Reference inversion. Base-frame prompts change the surface token and physical goal together; the reference-inversion control instead holds the physical relation fixed and changes its linguistic frame. The base-frame prompts are “Put the Rubik’s cube to the left of the bowl.” and “Put the Rubik’s cube to the right of the bowl.” The inverted prompts are “Place the Rubik’s cube so that the bowl is to the right of the Rubik’s cube.” and the same sentence with *left*. All four name the cube as the object to place, so inversion changes the frame of the relational clause and not the manipulandum; the direction word flips while the intended physical relation does not. The scorer, and the sign of M for this control, follow the physical goal rather than the prompt text. The $\pi_{0.5}$ study uses 341 resets in one symmetric DROID/RoboLab scene, with both forms

of both physical goals sharing reset and sampling state, for 1,364 episodes on seeds disjoint from every scene-intervention cohort.

Geometry interventions. Reflection is a pure world-space object-position intervention fixed in advance: it mirrors the centers of all movable objects about the robot sagittal plane without mirroring observations or un-mirroring actions. The robot, cameras, fixed scene geometry, prompts, seeds, and controller remain unchanged. We pair it with two further interventions fixed in advance: a seven-level sweep that moves only the reference object laterally and a symmetric-scene cohort that drives a registered movable-object asymmetry metric to zero (App. A). The symmetric package also changes which other objects are present, so it is not a position-only intervention and does not imply bilateral embodiment symmetry. Scene-effect claims are made only for rows that pass the prespecified endpoint-redirection positive control, which asks whether the policy still moves its endpoint in the commanded direction. Failed rows are reported but do not support a claim that the intervention attenuates a directional gap.

Scoring and inference. We do not relax the original task predicate. Each episode receives exactly one outcome label, assigned in a fixed order: pick failed, transport failed, wrong side, release failed, or correct. Every behavioral failure stays in its denominator. Marginal binary rates use Wilson intervals, paired continuous contrasts use seed-clustered percentile bootstraps with 20,000 resamples and exact sign tests, and layout interactions use exact within-seed label permutations. Absence claims require two one-sided equivalence tests (TOST) against prespecified margins, rather than nonsignificance [30]–[32]. Prompts, seeds, success tests, target quantities, margins, and the failure-label order are all fixed before the runs they affect. The public audit archive¹ records the available manifests, seeds, checkpoint revisions, and freeze times for the reported cohorts.

IV. REFLECTION REVERSES DIRECTIONAL COMPLETION

Under the prespecified reflection of Sec. III, the $\pi_{0.5}$ completion advantage reverses from RIGHT to LEFT. The paired outcomes are nearly mirrored: control RIGHT and reflected LEFT each succeed in 25/27 trials, whereas control LEFT and reflected RIGHT succeed in 4/27 and 9/27. The latter five-success difference is the part that reflection does not remove.

The eight-row Phase-A screen provides descriptive context (Fig. 2). $\pi_{0.5}$ and DreamZero favor RIGHT, Nano is near ceiling, and GR00T completes only 3/27 LEFT and 0/27 RIGHT trials; 49/54 GR00T episodes fail at pickup. The RoboTwin screening gaps are descriptively small, but two checkpoints later fail the endpoint-redirection positive control, so those rows do not test the layout-dependence claim.

A. Pure object-position reflection

Reflecting the centers of movable objects about the sagittal plane (Sec. III) gives four conditions per checkpoint (two directions by two layouts) over 27 seeds, or 108 episodes for

each of three DROID/RoboLab checkpoints. The $\pi_{0.5}$ reflection cohort uses seeds disjoint from the screen, which is why Fig. 2 reports 5/27 and 24/27 where the control condition below has 4/27 and 25/27. The primary outcome is the within-seed layout interaction in requested-side depth; success is secondary because binary outcomes can saturate by layout and direction. Nano, the checkpoint closest to the binary ceiling, illustrates why in Fig. 3.

The reflected-minus-control requested-side-depth interaction is negative in every reflection checkpoint (Fig. 1). The interaction is -34.6 cm for $\pi_{0.5}$ (95% CI $[-41.4, -28.5]$; 27/27 seeds; $p = 1.49 \times 10^{-8}$). It is -24.8 cm for Nano ($[-32.4, -17.3]$; 24/27; $p = 4.92 \times 10^{-5}$) and -14.1 cm for DreamZero ($[-19.9, -8.2]$; 23/27; $p = 3.11 \times 10^{-4}$). For $\pi_{0.5}$, reflection changes requested-side depth by -34.6 cm but endpoint redirection by -1.12 cm; for Nano, the corresponding changes are -24.8 and $+0.5$ cm. The two redirection interactions each have $p = .701$ because their signs split 12/15 and 15/12 over the same 27 seeds. Equivalence was not established, so neither redirection interaction is described as invariant.

Each seed contributes $(S_i^{R, \text{refl}} - S_i^{L, \text{refl}}) - (S_i^{R, \text{ctrl}} - S_i^{L, \text{ctrl}})$, so a negative value means reflection reduces or reverses the control-layout RIGHT completion advantage. The mean of this per-seed completion interaction is -1.37 on its $[-2, 2]$ scale ($p = 5.96 \times 10^{-8}$). In this $\pi_{0.5}$ DROID/RoboLab cohort, reflecting the movable-object centers flips the observed completion advantage under the unchanged frozen predicate. The reversal is also not an artifact of one policy draw: eight samples in each of 27 fixed scenes yield 41/216 LEFT and 197/216 RIGHT successes.

The reversal cannot be explained by a fixed directional marginal alone. Because the checkpoint is unchanged across layouts, a constant exposure-driven preference supplies no mechanism for the observed layout interaction. Exposure conditioned on object configuration remains consistent with the result, however, and these experiments do not distinguish geometry-as-physics from geometry-as-coverage. Section VIII states the experiment needed to separate them.

B. A graded intervention and converging controls

Reflection is a large intervention, so we also move the bowl across seven lateral positions while retaining the same Nano checkpoint and 15 matched seeds per level (210 episodes). Across all seven registered positions, the requested-depth contrast has a slope of 1.12 cm of depth per cm of reference displacement (95% CI $[0.72, 1.56]$); 13 of 15 seed-level slopes are positive (exact $p = .00739$). A post hoc middle-five sensitivity is weaker: 0.49 (95% CI $[-0.04, 0.97]$), with 11 of 15 positive slopes ($p = .118$). We therefore claim a graded dependence across the seven tested positions, not an interior linear law. Because the fitted zero crossing lies outside the tested support, the data do not identify a balanced interior layout.

Fig. 4 also shows why binary success alone is a weak diagnostic here: LEFT completion changes from 15/15 to 10/15, while RIGHT stays close to ceiling. Separate registered Nano cohorts

¹<https://github.com/adeeb10abbas/steerable>

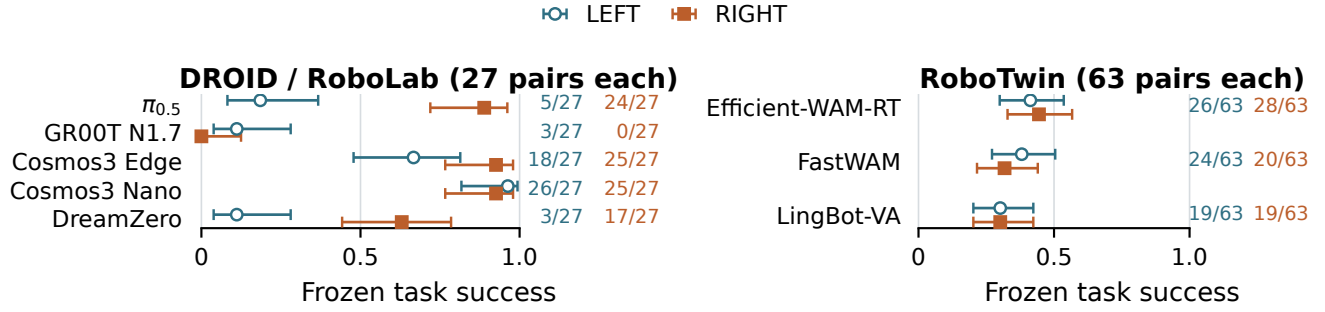


Fig. 2. **Directional task success varies across the eight checkpoint-arena rows.** Points show marginal LEFT and RIGHT success with 95% Wilson intervals. DROID/RoboLab uses 27 matched pairs per row and RoboTwin 63; their tasks and frozen predicates differ. The screen is descriptive, and every causal comparison in this paper is within checkpoint; the figure does not support cross-model or cross-arena ranking.

support the same layout-dependence claim without being pooled with reflection. For Nano, the RIGHT-minus-LEFT success gap changes from $(22 - 26)/27 = -4/27$ when the target starts robot-left of the reference to $(27 - 23)/27 = +4/27$ when it starts robot-right. Under the registered right-start-minus-left-start convention, the interaction is therefore $+8/27 = +0.296$ ($p = .0156$). Also for Nano, swapping target and reference roles changes the depth contrast by $+13$ cm (95% CI $[6, 21]$, $p = .00151$). The role swap keeps the direction word and relation template fixed but changes the manipulandum and physical goal; reference inversion instead keeps both fixed while changing the linguistic frame.

V. REFERENCE INVERSION COLLAPSES ENDPOINT SEPARATION

Reference inversion collapses endpoint separation for $\pi_{0.5}$ from 23.28 cm to 0.19 cm on matched physical goals (Fig. 5). This is the linguistic counterpart to the scene controls: familiar base-frame endpoint separation does not survive when the same relation is expressed from the other object’s perspective.

Matched reference inversion. Reference inversion asks whether the same physical relation survives being expressed from the other object’s frame. Under inversion the direction word and the physical goal point opposite ways, which separates two behaviors that base-frame prompts cannot tell apart. If the policy tracks the relation, endpoint position for the leftward goal minus that for the rightward goal should be positive. If it tracks the direction word regardless of the sentence frame, the contrast should be negative, reaching -23.28 cm if the base-frame response transferred unchanged.

Base-frame prompts separate the endpoints by $+23.28$ cm (95% CI $[21.89, 24.56]$), whereas reference-inverted prompts yield only $+0.19$ cm ($[-0.88, 1.27]$). The near-zero response supports neither relation-preserving transfer ($+23.28$ cm) nor simple token mirroring (-23.28 cm); its interval includes zero, although equivalence to zero was not established. Out-of-distribution phrasing remains an alternative explanation that this experiment does not separate from failed relation transfer. This conclusion is specific to endpoint direction, not all prompt sensitivity; all 682 base-frame/inverted physical-goal pairs execute different action traces.

Inversion also reduces placement depth within each physical goal. The inverse-minus-base-frame change is -9.78 cm for the leftward goal (90% CI $[-10.92, -8.59]$) and -13.31 cm for the rightward goal ($[-14.07, -12.54]$); both intervals lie outside the prespecified ± 4.15 cm equivalence margin. Success falls from 210/341 to 151/341 for the leftward goal and from 297/341 to 240/341 for the rightward goal. These rates make a wholesale switch to moving only the bowl less consistent with the evidence: the predicate scores placement of the cube, yet 151 and 240 of 341 inverted episodes still satisfy it. Because reference-object displacement was not measured, the counts do not exclude incidental bowl motion or a mixed strategy. The corresponding changes, -17.30 and -16.72 percentage points, do not meet the prespecified ± 15.56 -point equivalence criterion. Base-frame response is therefore insufficient evidence that a physical relation is preserved across reference frames.

Base-frame and fixed-observation checks. All 324 base-frame minimal pairs across the eight rows execute different action sequences: 135 DROID/RoboLab pairs and 189 RoboTwin pairs. Endpoint ordering is positive in 125/135 and 128/189 pairs, respectively. These bitwise differences supply the matched trace-difference positive control; they do not show that the policy represents the relation independently of its wording or guarantee task completion.

The fixed-observation diagnostic of Sec. III separates prompt response from sampling noise. For $\pi_{0.5}$, the prompt-induced median action difference is 4.1% of the pooled same-prompt cross-seed median in control and 3.3% after reflection, yet its difference vectors share a consistent orientation across seeds (mean cosine 0.540 and 0.652; permutation $p < 10^{-5}$). For Nano, the corresponding ratios are 0.720 and 1.192, but the orientations are far less consistent (mean cosine 0.027 and 0.079; permutation $p = .167$ and $.00314$). For DreamZero, all 702 same-prompt cross-seed distances are zero in each layout, while median prompt RMS is 0.00923 in control and 0.02348 after reflection. Its ratio is therefore undefined rather than small; whether the zero measured variation belongs to the checkpoint or to harness seeding remains unresolved. These open-loop rankings do not match closed-loop endpoint ordering, so action difference is a positive control, not a standalone grounding

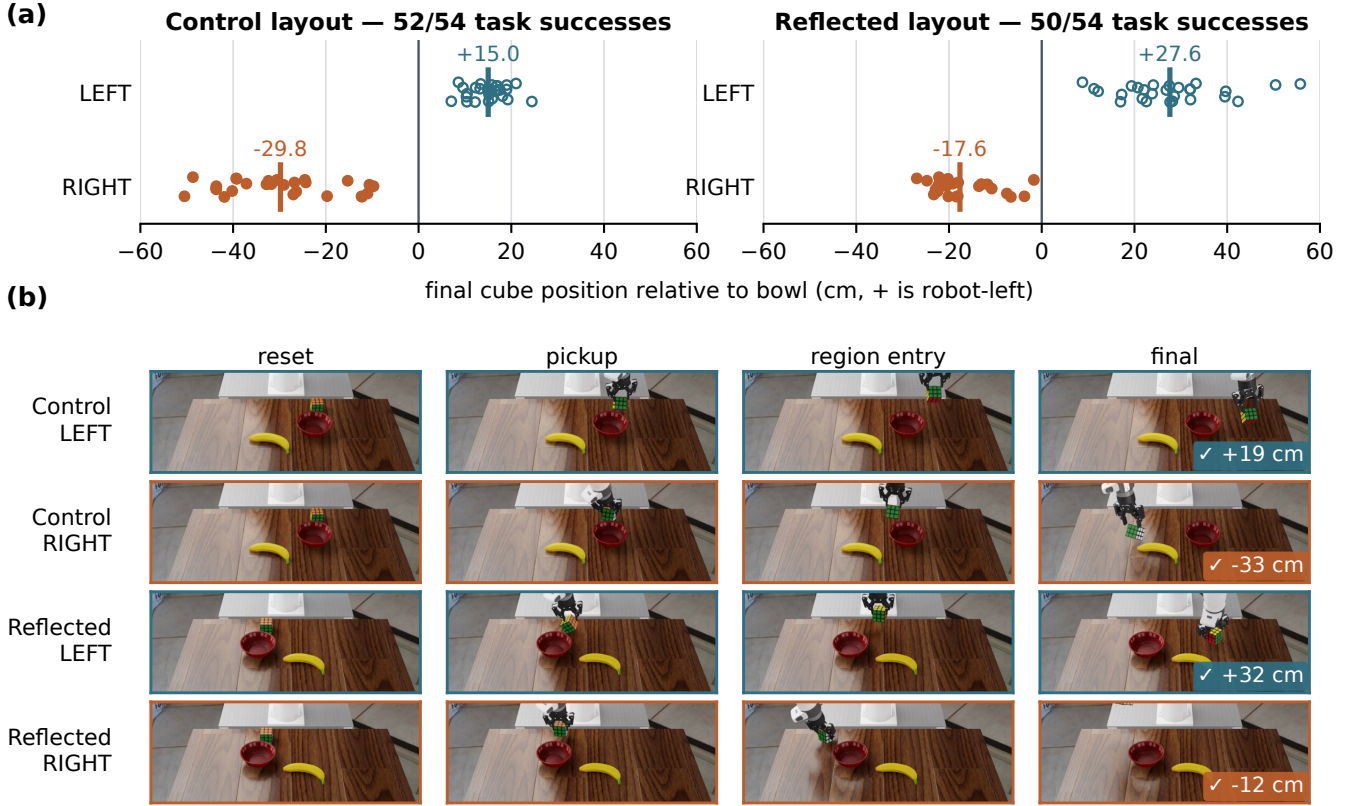


Fig. 3. **Reflecting movable-object positions moves the placement while the binary score stays at ceiling (Nano, 108 episodes).** (a) Final lateral cube position for every seed in the four conditions; bars mark their means and titles give task completion. The prompts still separate after reflection, by 45.3 cm against 44.8 cm in the control layout, but both distributions shift by about 12 cm, which is the requested-side depth interaction of -24.8 cm reported below. Completion moves only from 52/54 to 50/54. (b) The executed rollout for the lowest released seed in each condition, sampled at that episode’s logged events, with ticks marking the frozen predicate and numbers repeating the final signed offset. Robot-left appears toward the image right in this viewport, so the signed values rather than the images fix direction. Rollouts were selected by lowest released seed before any outcome was read.

score.

VI. BINARY CEILINGS HIDE PLACEMENT CHANGE

Nano supplies the binary-ceiling case previewed in the abstract. Its completion interaction is zero ($p = 1$), whereas requested-side depth shifts by -13.9 cm ($p = 1.8 \times 10^{-4}$). The zero binary interaction therefore conceals a substantial placement change.

The symmetric-scene cohort contains 4,096 episodes (2,048 matched pairs) across five checkpoint–arena rows. The dense rows contribute 763 $\pi_{0.5}$ pairs and 1,123 Nano pairs; Edge, DreamZero, and FastWAM contribute 54 each. The intervention zeros the registered movable-object asymmetry metric but replaces a scene package rather than changing positions alone. Four DROID/RoboLab checkpoints pass the endpoint-redirection control and support a scene-effect interpretation; FastWAM and a separate 108-episode LingBot-VA cohort fail it and are reported without that interpretation. GR00T is excluded because it almost never completes, and the $\pi_{0.5}$ seeds are disjoint from those used for reference inversion.

For $\pi_{0.5}$, 341 prompt pairs at each endpoint show the binary RIGHT-minus-LEFT gap falling from 74.2 to 20.2

percentage points and the requested-depth gap from 17.0 to 5.75 cm. At the symmetric endpoint, their paired 90% intervals are $[15.0, 25.2]$ points against a prespecified ± 15.56 -point margin and $[4.85, 6.66]$ cm against ± 4.15 cm; neither meets its equivalence criterion. The prespecified power audit at $n = 341$ gives an 80%-power minimum detectable effect of 7.8 percentage points for the binary gap, relative to the 15.56-point equivalence margin. A separately registered matched core uses 27 seeds shared across layouts; it estimates interactions of -51.9 points (95% CI $[-81.5, -22.2]$, $p = .00786$) and -12.4 cm (95% CI $[-19.7, -4.9]$, $p = .00348$), respectively. Base-frame redirection at the symmetric endpoint is $+23.9$ cm, with 335/341 positive pairs.

The other positive-control-passing checkpoints show the effect’s cross-checkpoint heterogeneity (Fig. 4). DreamZero’s observed binary gap changes from $+51.9$ to -3.7 points (interaction -55.6 points, $p = .00085$), bringing the measured gap near zero without establishing equivalence. Edge is underpowered for its -14.8 -point binary interaction, whose 95% interval $[-40.7, +7.4]$ includes zero, while its depth interaction is -19.2 cm (95% CI $[-26.0, -11.5]$).

Failure labels also change descriptively across the full $\pi_{0.5}$

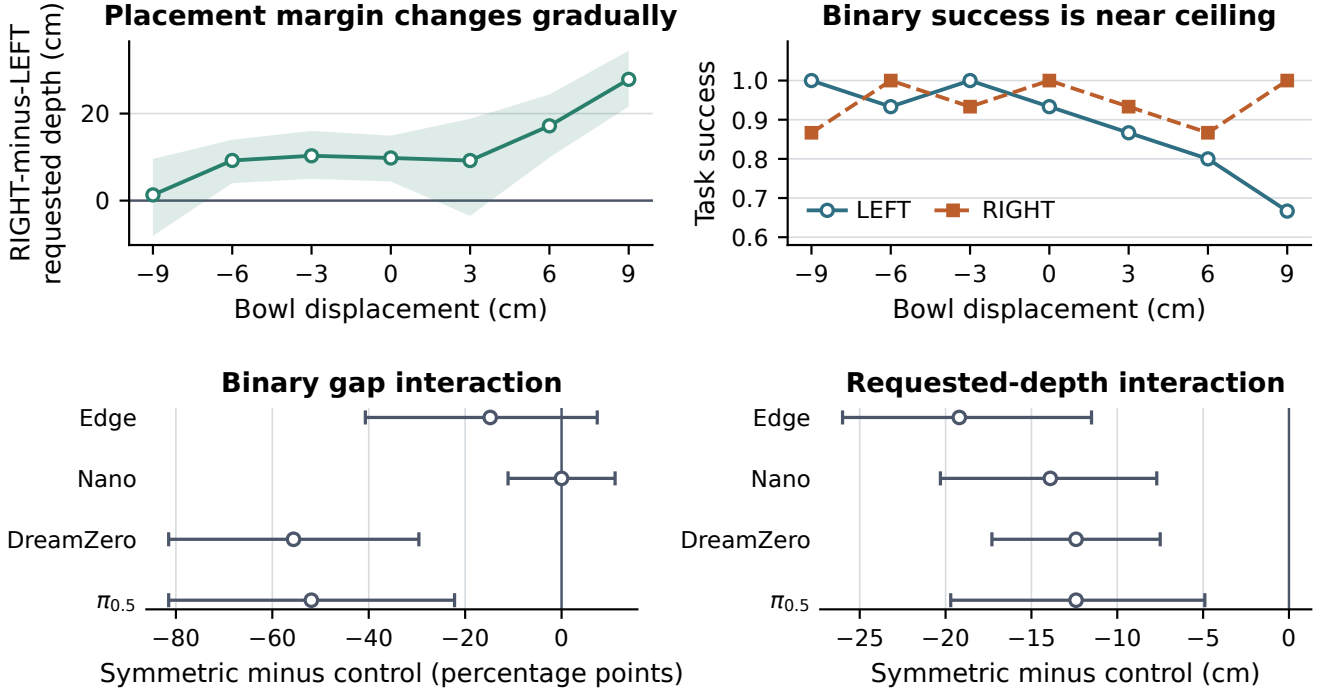


Fig. 4. **Two scene interventions, measured continuously and binarily.** Top: for Nano, the directional placement contrast across the seven-position reference-object sweep, with mean RIGHT-minus-LEFT requested-side depth and seed-clustered 95% intervals on the left and frozen task success by requested direction on the right; each position contains 15 matched seeds. Bottom: symmetric-minus-control interactions in the matched 27-seed core of the positive-control-passing DROID/RoboLab cohort, with seed-clustered 95% intervals; requested-depth disparity decreases in every checkpoint. In both interventions the continuous measure exposes variation that binary success compresses near ceiling, as for Nano. Every comparison is within checkpoint; the panels do not support cross-model ranking.

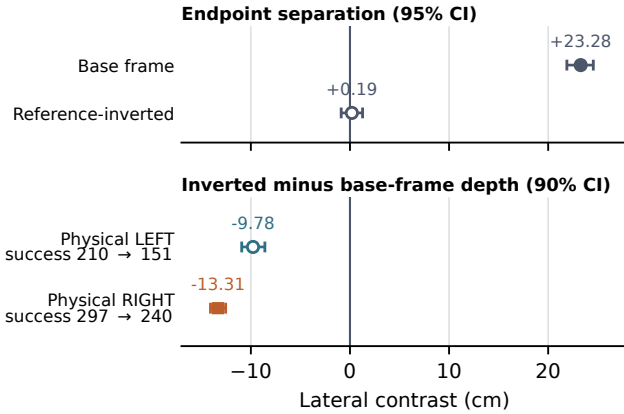


Fig. 5. **Reference inversion collapses directional endpoint separation for $\pi_{0.5}$.** Top: leftward-minus-rightward physical-goal endpoint separation with seed-clustered 95% bootstrap intervals; positive is relation-consistent. Bottom: inverted-minus-base-frame requested-side depth with 90% intervals; labels give base-frame-to-inverted success counts out of 341. This cohort’s seeds are disjoint from those of the scene-intervention experiments.

endpoint cohorts, pooling directions within 682 episodes per layout. The control endpoint has 1 pick, 267 transport, 41 wrong-side, and 2 release failures; the symmetric endpoint has 0, 158, 7, and 0, respectively, for 311 versus 165 total failures. These unpaired endpoint totals are descriptive bookkeeping, not a separate interaction test.

VII. DISCUSSION

A binary score can fail in opposite directions. Reflection shows how layout-dependent completion can be misread as a grounding failure; reference inversion shows how a reassuring familiar-frame response can be misread as relation-level grounding. The four measurements locate where these interpretations fail: a binary ceiling can hide a placement change, and an executed trace can change without expressing the intended relation. The frozen predicate remains the final completion test, but it cannot identify the cause.

A directional gap belongs to the policy-layout pair. Across matched scene interventions, the directional contrast changes while the policy, prompts, and seeds remain fixed, so it is not a layout-invariant checkpoint property. The residual that survives the symmetric scene also prevents attributing the entire gap to layout: the controls diagnose dependence without identifying its remaining cause.

A minimum directional-grounding protocol. For relational placement, we recommend matched base-frame LEFT/RIGHT prompts under identical reset and policy seeds, paired with action distinctness, a signed endpoint measure, requested-side depth, and the frozen success predicate. Where stochasticity is material, add same-prompt repeats. Attribution further requires at least one matched scene intervention with a preserved endpoint-redirection positive control and a reference inver-

sion that preserves the physical goal. Claims that an effect disappeared should rest on equivalence tests rather than a nonsignificant difference.

VIII. LIMITATIONS

Our design trades external validity for control. Exact state matching needs a resettable simulator, so hardware contact uncertainty and natural variation in human instructions remain untested. The object-layout conclusions hold only for the DROID/RoboLab rows that pass the endpoint-redirection control; the RoboTwin rows screen behavior, not a cross-arena replication or refutation. The task family is narrow as well: two named objects, one lateral relation, and one reference inversion on one $\pi_{0.5}$ checkpoint.

The interventions establish scene dependence but do not distinguish physical reachability from training coverage correlated with similar object configurations. As Sec. IV explains, reflection rules out a fixed directional marginal as the complete explanation; it does not rule out configuration-conditioned exposure. The evaluated releases disclose neither episode membership, sampling weights, training layouts, nor configuration-conditioned captions, and corpus-wide LEFT/RIGHT counts would not recover the exposure relevant to these scenes. That attribution is therefore not resolvable for these released checkpoints. DreamZero’s public demonstrations do include lateral placements referring to a bowl, so our task may sit closer to its demonstrated set than to the other rows’. Two measurement gaps remain: we did not track reference-object displacement under inversion, though inverted success rates show that the cube still satisfied the placement predicate (Sec. V), and the fixed-observation diagnostic is action-space only. Settling the exposure question requires training policies from scratch on corpora with controlled LEFT/RIGHT balance within matched object configurations and then rerunning the same interventions; released checkpoints cannot provide that test.

IX. CONCLUSION

Matched controls expose three distinct failures of attribution. Reflection makes the same checkpoint favor the opposite completion direction; reference inversion collapses familiar-frame endpoint separation on matched physical goals; and Nano’s binary ceiling conceals a large placement change. Together, these results show that scene layout and linguistic frame change the directional score, while Nano shows that a binary task boundary can conceal continuous placement change. Directional success therefore does not isolate language grounding.

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Short name	Released artifact identifier and revision
$\pi_{0.5}$	pi05_droid_jointpos_polaris v2a010-manifest-f5a56d9565f9381cc
GR00T	nvidia/GR00T-N1.7-DROID 05e7cc97e40dbd33b0890c35cc0214fcb0547ab5
Edge	nvidia/Cosmos3-Edge-Policy-DROID 3ea407af3e156c0af3b4bb6edd85842cc9a58777
Nano	nvidia/Cosmos3-Nano-Policy-DROID 6706d7680581c255ff61e0f3bb49d90eac55c79e
DreamZero	GEAR-Dreams/DreamZero-DROID 96ad344138c66e82536422432ad742f015784942
Efficient-WAM-RT	jiajun0613/Efficient-WAM_RoboTwin/ Efficient-WAM-RT 81280a79e8ac69dd6ffbb9ce8698e00d122ec07fd
FastWAM	yuanty/fastwam/ robotwin_uncond_3cam_384.pt 139eebb6d90cdd9bdbbe465f72c6edc9ad5a518a
LingBot-VA	lerobot/lingbot_va_robotwin dle1f93a84eaf9bca9880856fda800cc98cc8eaa robbyant/lingbot-va-posttrain-robotwin 8c9dea8abbc5c91cc9e18bc3264b8915083bbe70

The $\pi_{0.5}$ row is an internal release rather than a public repository revision, so its recorded identifier is a manifest digest, abbreviated here and given in full in the released manifest. LingBot-VA is evaluated as a base checkpoint plus a RoboTwin post-training checkpoint, so both revisions are required to reproduce it.

APPENDIX A SYMMETRIC-SCENE ASYMMETRY METRIC

The registered metric is the root-sum-square departure of the movable objects from bilateral symmetry about the robot midline. It measures midline objects from the midline itself and mirror-paired objects from each other’s reflection, with position weighted at 10 per metre and orientation at 1 per radian. The control layout scores 3.85 and the symmetric layout 0.00, within prespecified millimetre and half-degree tolerances. The robot, cameras, wrist mounting, reset posture, prompts, seeds, and controller are unchanged across the two scenes, but the companion-object inventory changes; the metric therefore certifies movable-object symmetry rather than full-scene or embodiment symmetry.

APPENDIX B EVALUATED CHECKPOINT PROVENANCE

Each row below lists the recorded artifact identifier and revision used in the evaluation. The public manifests provide the available content hashes and runtime metadata. They do not expose training-episode membership, sampling weights, or configuration-conditioned caption exposure.